

Active Directory Home Lab – Windows Server 2022

Overview

I set up a Windows Server 2022 Active Directory environment in a virtual machine in this home lab to see how identity and access management works in a real business network.

In the lab, there was a Windows Server 2022 domain controller and a Windows 11 Pro client machine that was part of the domain. This let me see how businesses manage users, devices, and security policies across many systems from one place.

The goal of this project was to get some real-world experience with:

- Active Directory Domain Services
- Organizational Unit design
- User and group management
- Group Policy Objects
- Network file sharing
- Security policy implementation
- Domain-joined client management

Lab Environment

Virtualization Platform

VirtualBox

Server

Windows Server 2022

Configured as a Domain Controller

Client Machine

Windows 11 Pro

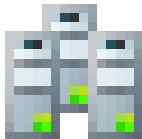
Joined to the Active Directory domain for testing policies and user authentication.

Core Services Implemented

- Active Directory Domain Services
- Group Policy Management
- File and Folder Sharing
- Security Policy Enforcement

Domain Setup

After installing Active Directory Domain Services, I created the domain:



Johncam.local

The Windows Server system acts as the domain controller, which manages:

- Authentication
- User accounts
- Security policies
- Device management

After that, the Windows 11 Pro machine was added to the domain. This meant that users who were made in Active Directory could use their domain credentials to log into the client machine.

Organizational Unit Structure

To simulate a real organization, I created organizational units based on geographic regions.

Regional OUs

- USA
- Europe
- Asia

I made organizational units based on departments within the USA OU.

Department OUs

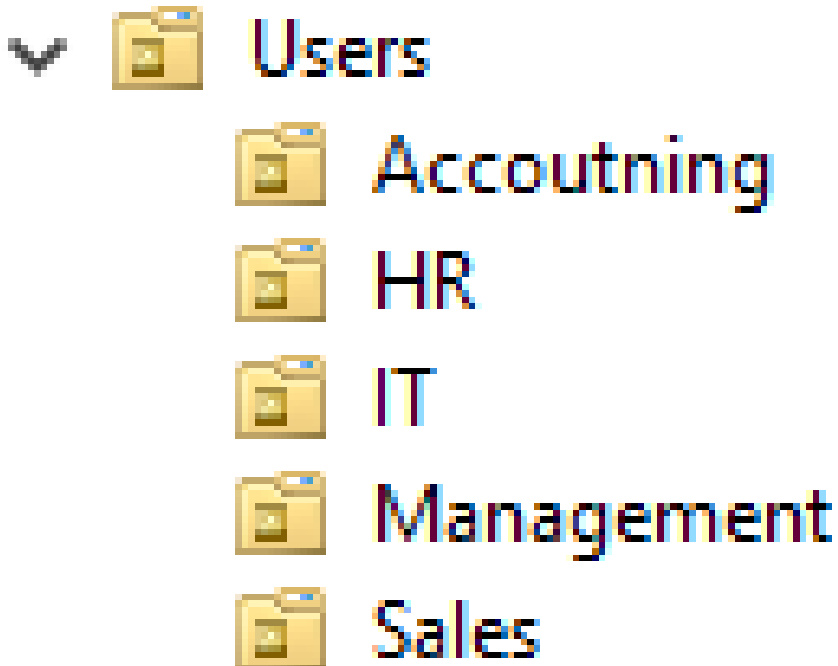
- IT
- Accounting
- HR
- Sales
- Management

This setup lets administrators set rules and permissions based on where people work or what department they work in.

Users, Groups, and Computers

User Accounts

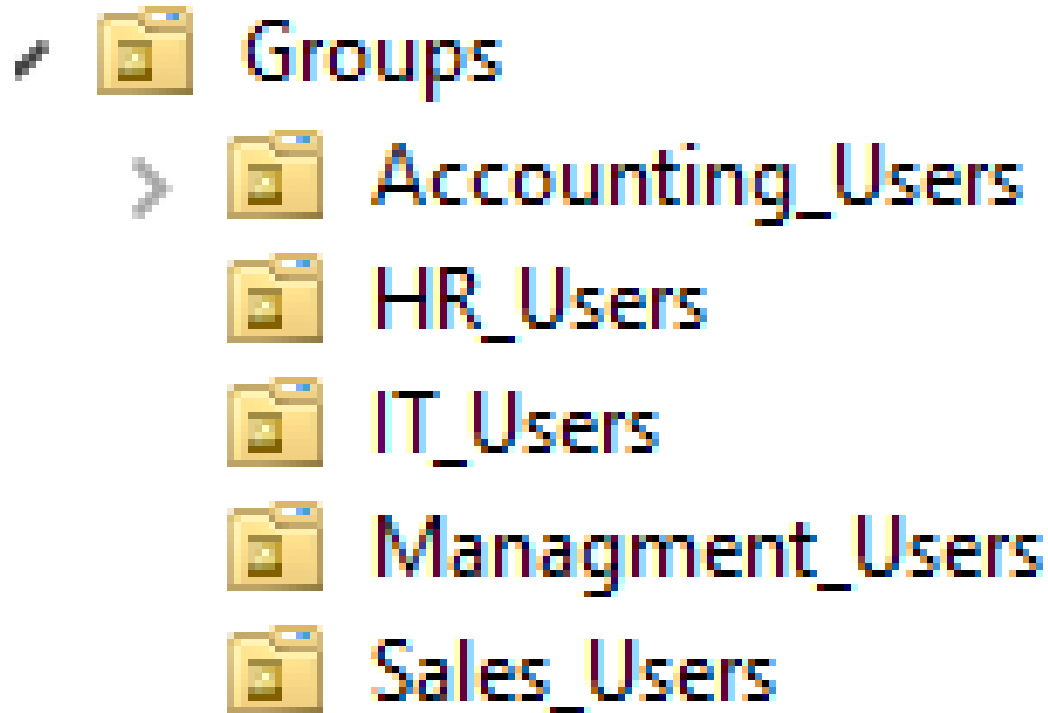
Within each department, I created multiple users to simulate employees.



Three users were created for each department to simulate multiple employees within the organization.

Security Groups

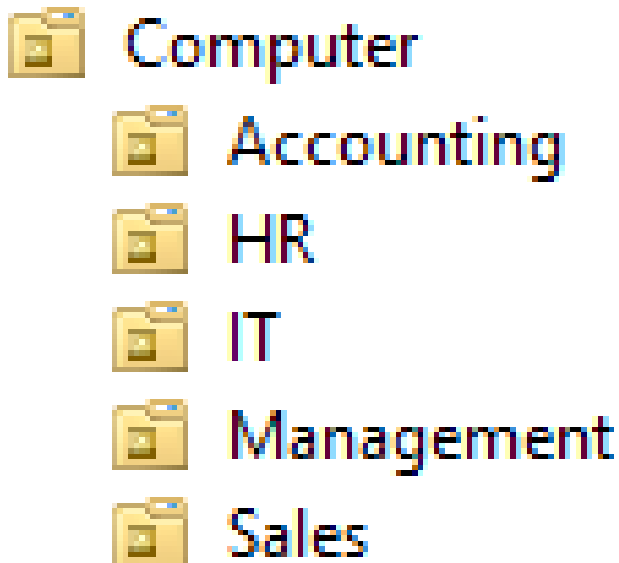
Security groups were created to simplify access control and permissions management.



In business settings, it's common to use security groups to give permissions to groups of users instead of just one person.

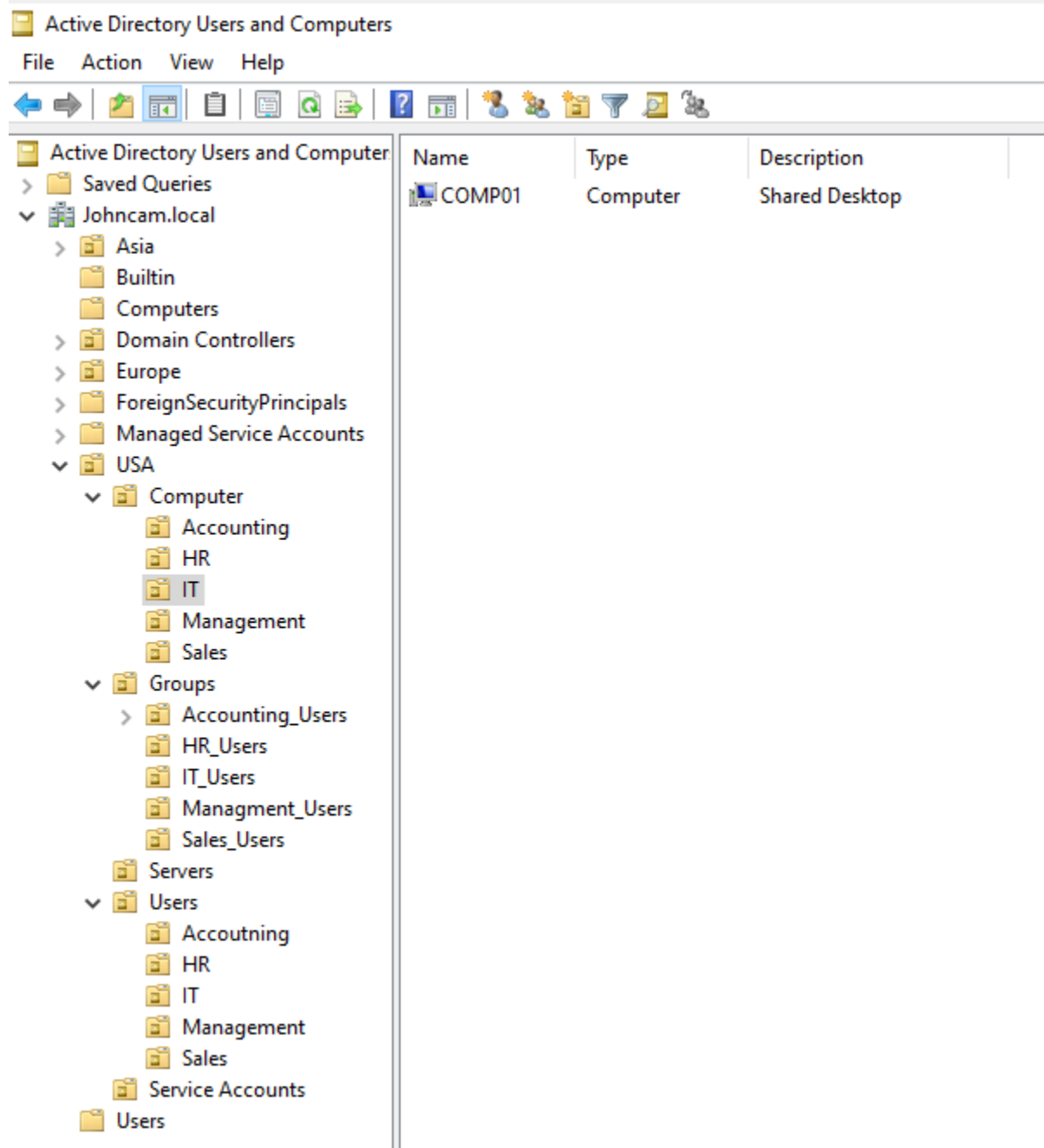
Computer Organization

Computers were also organized within Active Directory.



This allows policies to be applied specifically to devices within certain departments.

The Windows 11 Pro client machine was added to the domain and placed within the appropriate organizational unit.



Group Policy Implementation

The Group Policy Management Console was used to make and apply several Group Policy Objects to the organizational units.

These rules made it possible to centrally manage the settings for users and systems.

Policies Implemented

Password Policy

- Password complexity requirements
- Minimum password length
- Password expiration settings

Account Lockout Policy

- Account lockout after multiple failed login attempts

Device Restrictions

- Disabled USB storage devices

Desktop Configuration

- Enforced company desktop wallpaper

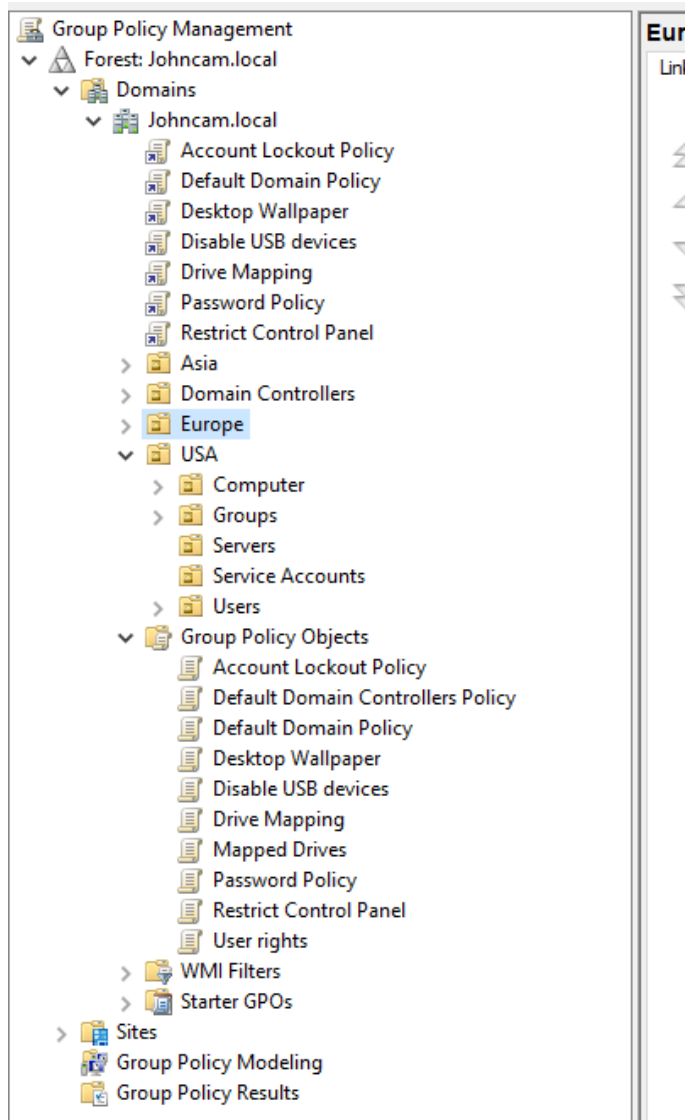
System Restrictions

- Restricted Control Panel access

Drive Mapping

- Automatically mapped network drives for users upon login

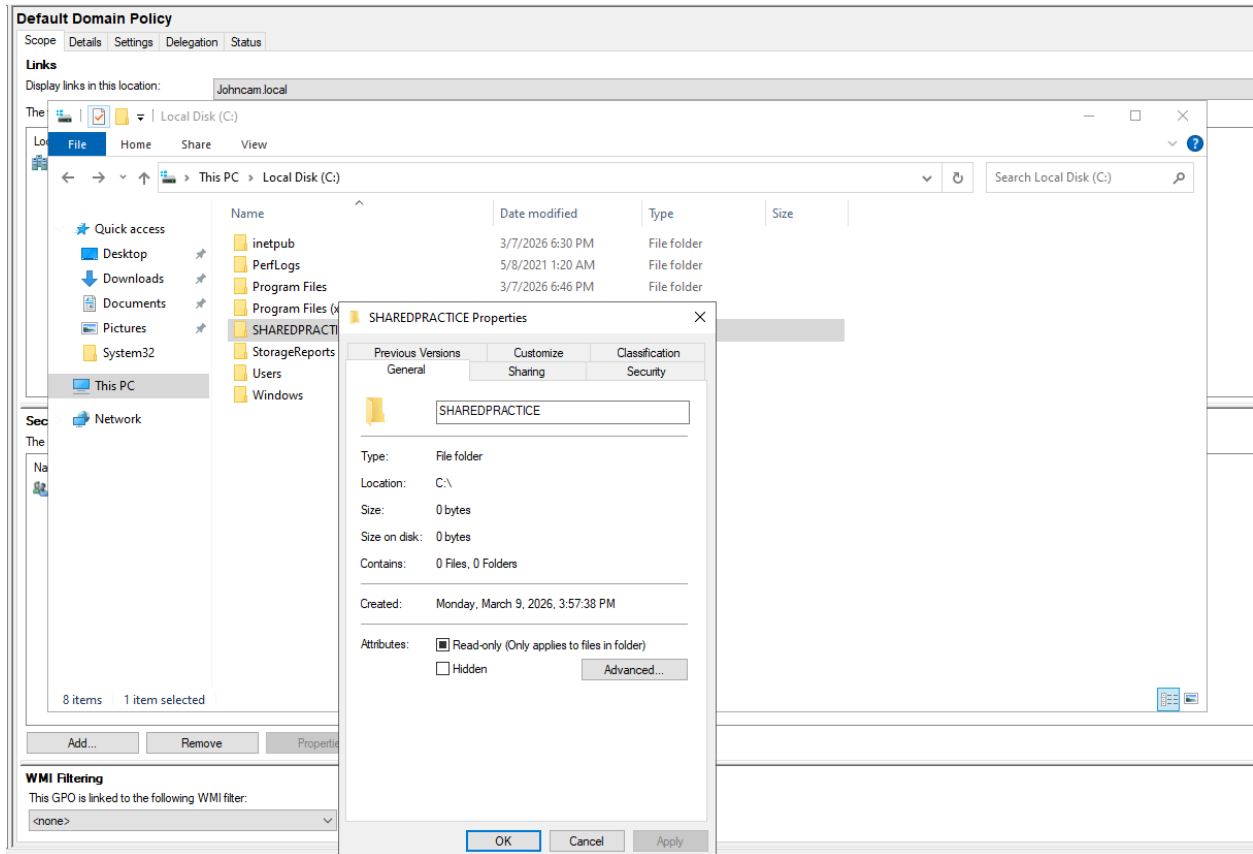
These policies were applied to specific organizational units so they would only affect the intended departments.



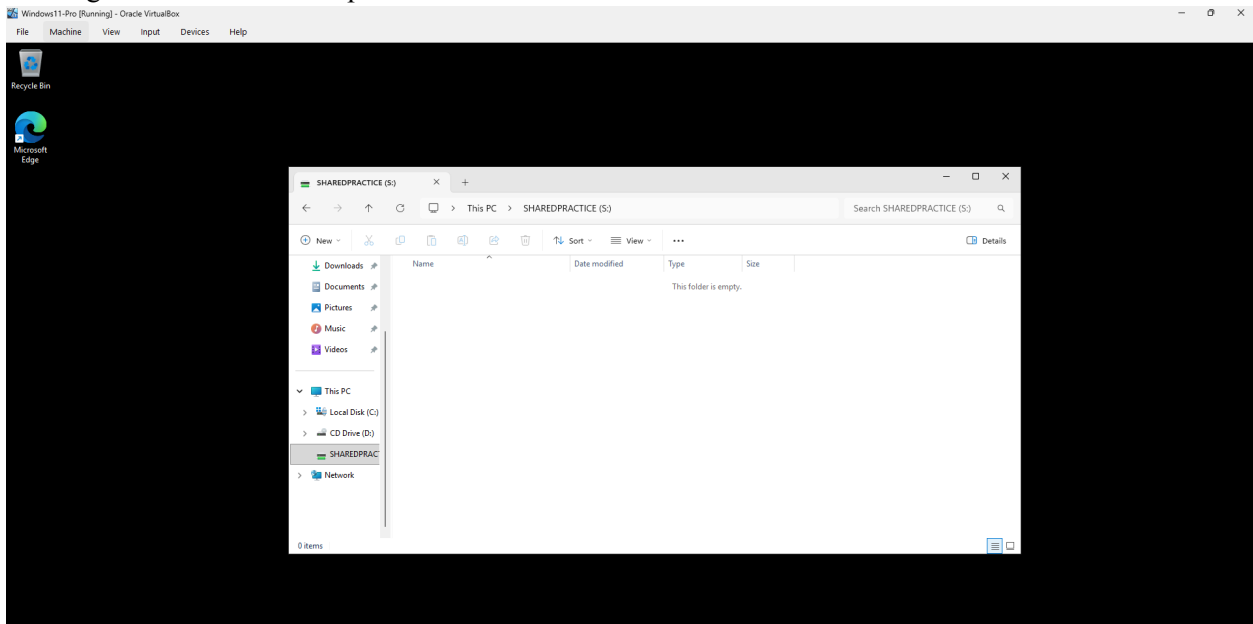
File Services and Network Sharing

To simulate a centralized company file server, shared folders were created on the Windows Server.

Shared Folder Structure



Looking at it from client computer



Permissions were assigned using Active Directory security groups, ensuring that users only had access to their department's files.

Network drives were then automatically mapped to the Windows 11 client machine using Group Policy.

Client Machine Testing (Windows 11 Pro)

The Windows 11 Pro client machine was used to test the Active Directory environment.

Testing included:

- Logging into Windows using domain accounts
- Verifying Group Policy enforcement
- Confirming network drives were mapped automatically
- Ensuring user access permissions were working correctly
- Testing account lockout policies

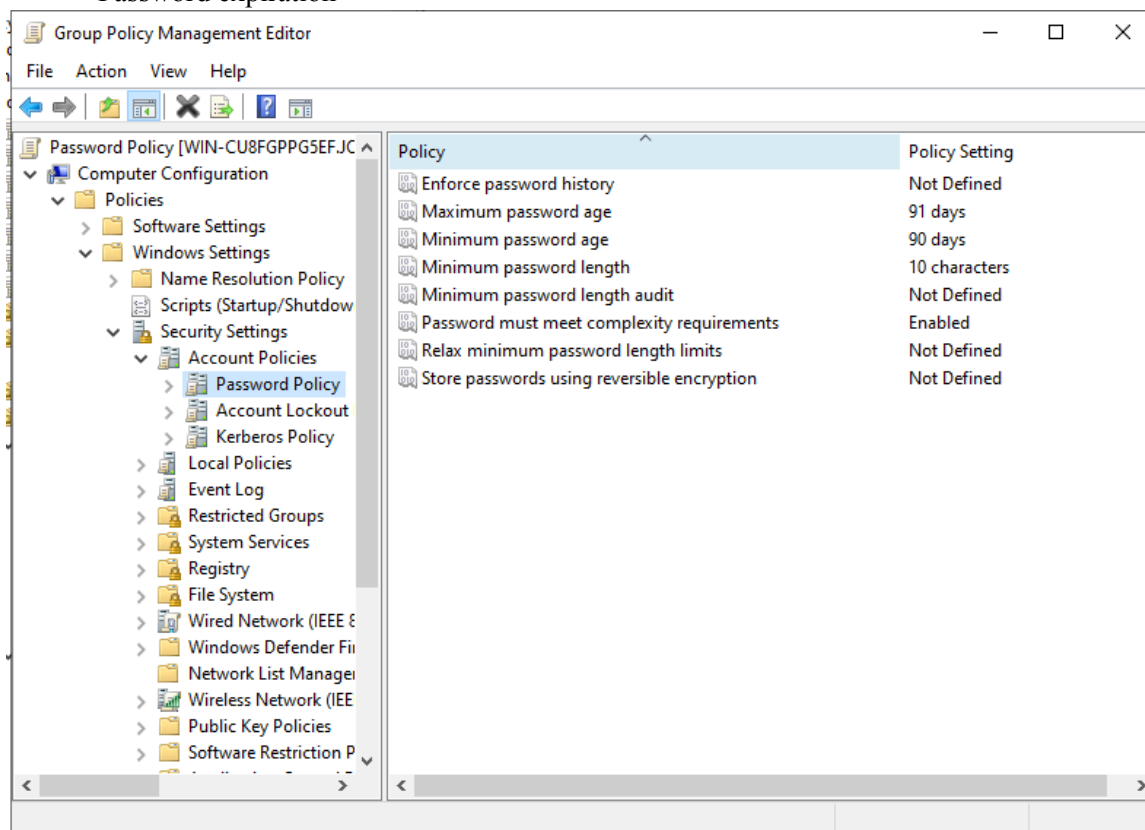
I was able to make sure that policies set up on the domain controller were successfully applied to devices that were connected to the domain.

Security Policies

Additional security policies were implemented to simulate enterprise security controls.

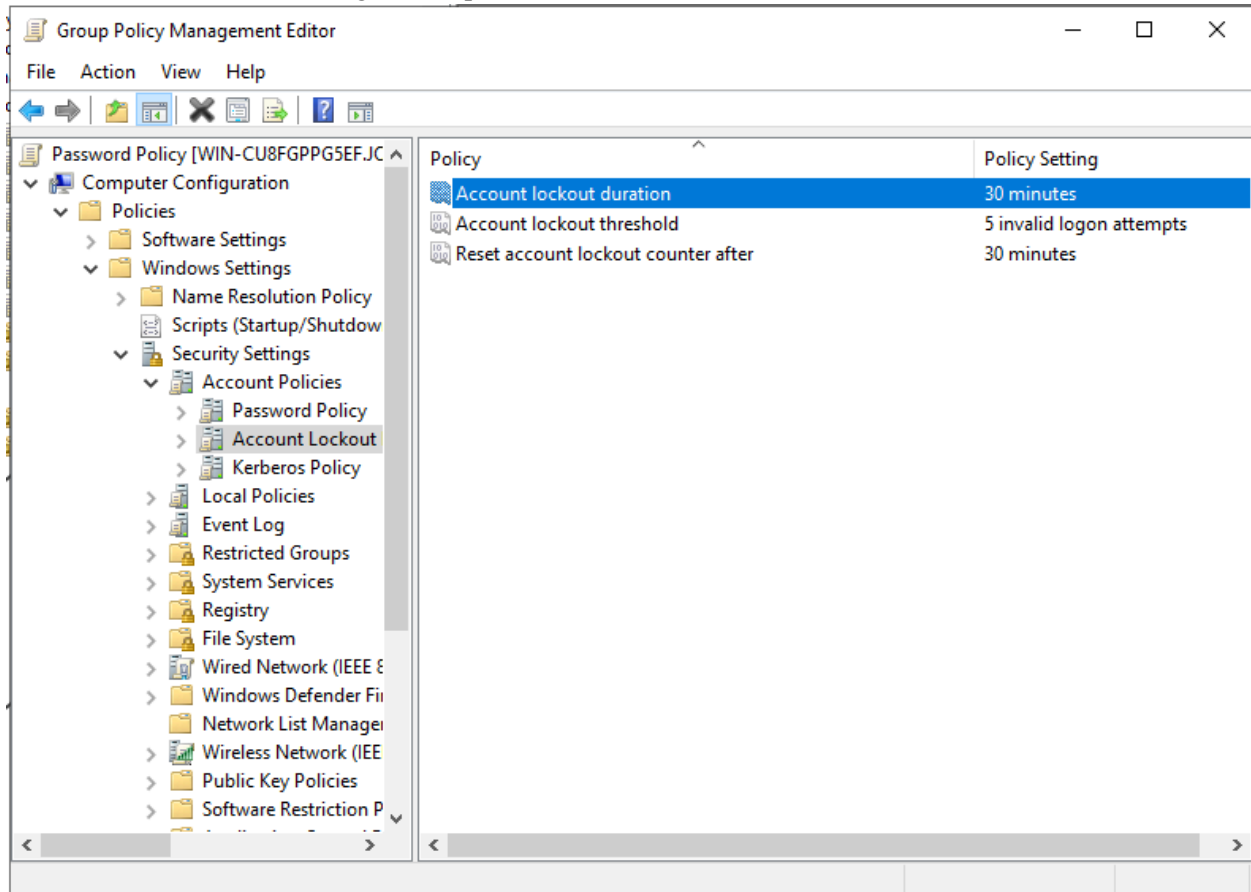
Password Security

- Minimum password length
- Complexity requirements
- Password expiration



Account Lockout

- Lock account after repeated failed login attempts
- Prevent brute-force login attempts



Fine-Grained Password Policies

Fine-grained password policies allow different password rules to be applied to different groups within the organization.

Key Skills Demonstrated

This lab demonstrates hands-on experience with:

- Active Directory Domain Services
- Organizational Unit design
- User and group management
- Group Policy configuration
- File server configuration
- Network drive mapping
- Enterprise security policy implementation
- Windows Server administration
- Domain client management
- Virtualized lab environments

Tools Used

- VirtualBox
- Windows Server 2022
- Windows 11 Pro
- Active Directory Domain Services
- Group Policy Management Console
- Windows File Services

What I Learned

With this project, I got to work with:

- Designing an Active Directory organizational structure
- Managing users, groups, and computers within a domain
- Implementing security policies using Group Policy
- Configuring shared network resources
- Testing domain authentication using a Windows 11 client system

This lab helped strengthen my understanding of enterprise identity and access management systems.